BOHR MODEL

General Chemistry I, Lecture Series 4

Qixi Mi

Reading:
OGB8 §§4.2, 4.3

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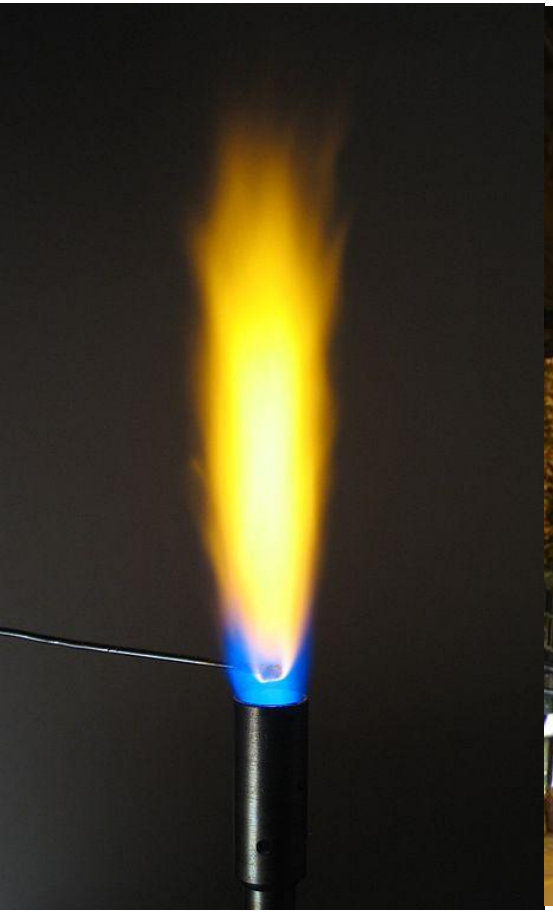
Outline

- Flame Tests and Spectra
- Balmer, Rydberg & Bohr
- The Bohr model

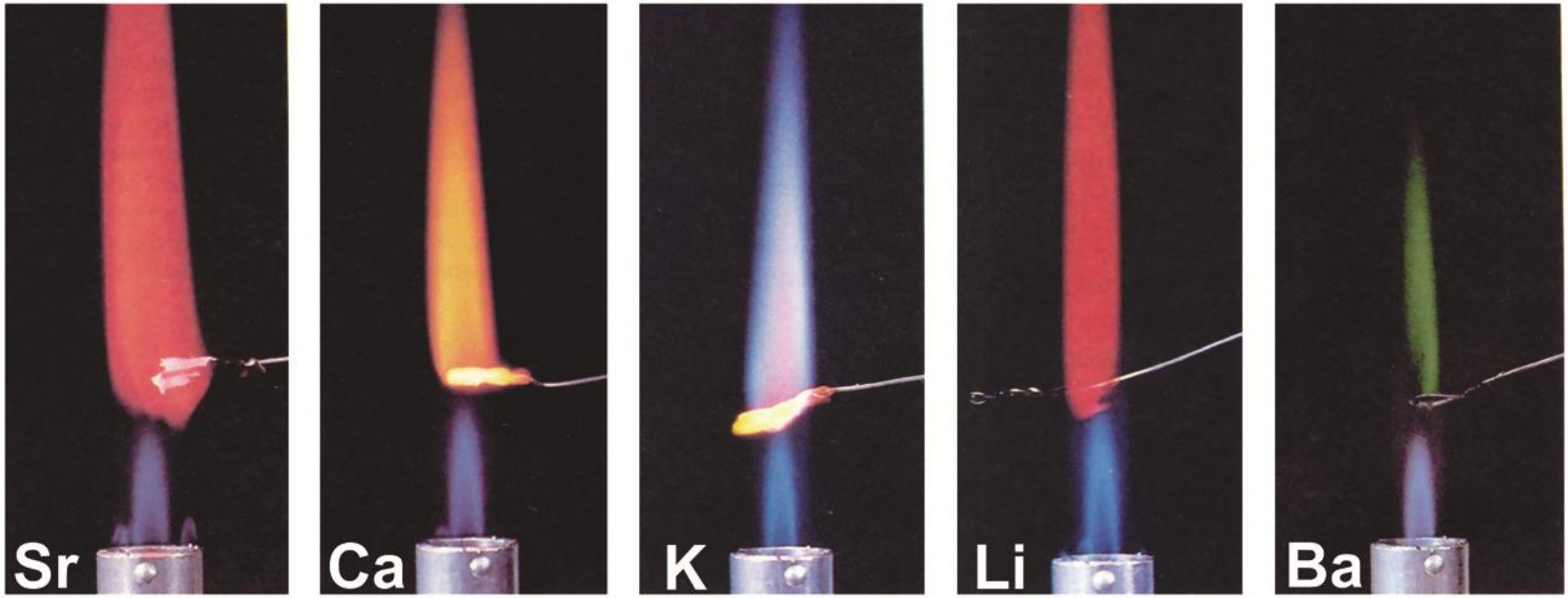
Sodium Flames and Sodium Lamps

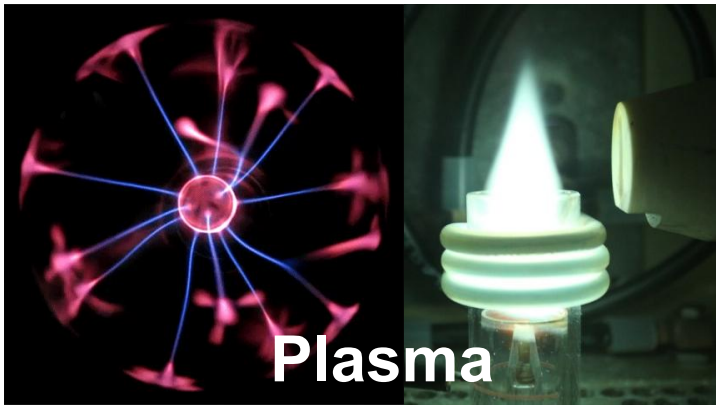


NaCl
Soda-lime glass
Sweat
…

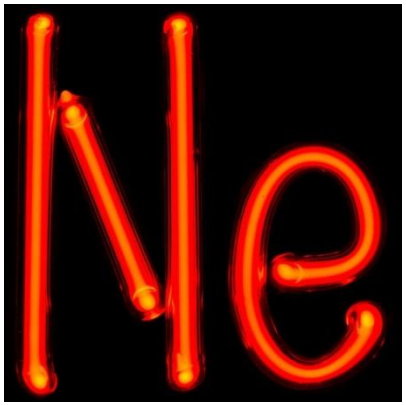
An element in various compounds shows the same flame color.

Flame Tests





Demo: Gas Discharge Tubes

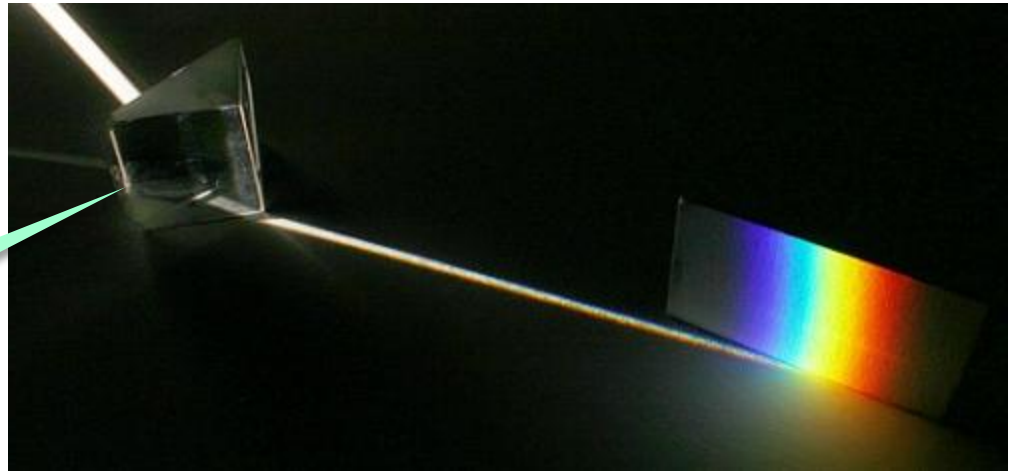


Neon lamp 氖(霓虹)灯

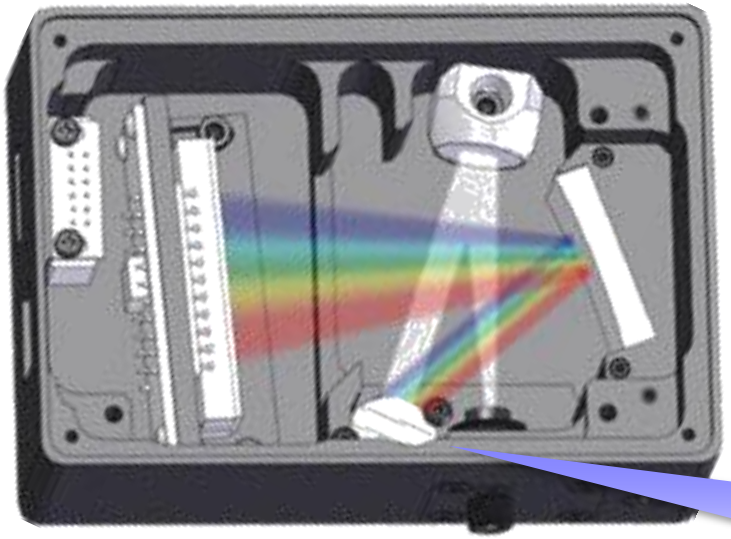
Xenon lamp 氙灯

Mercury lamp 汞灯

From Colors to Spectra

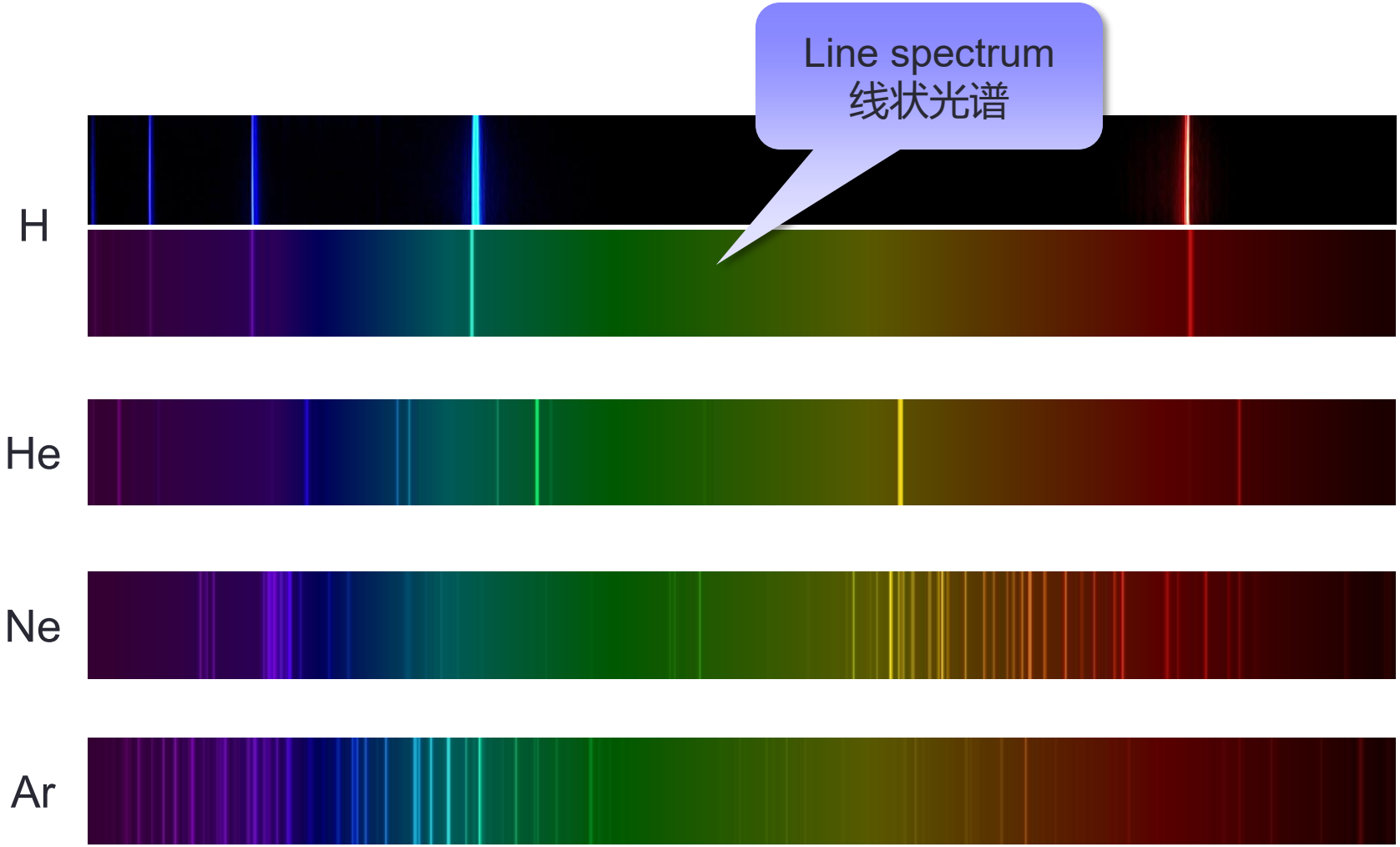


Prism 棱镜

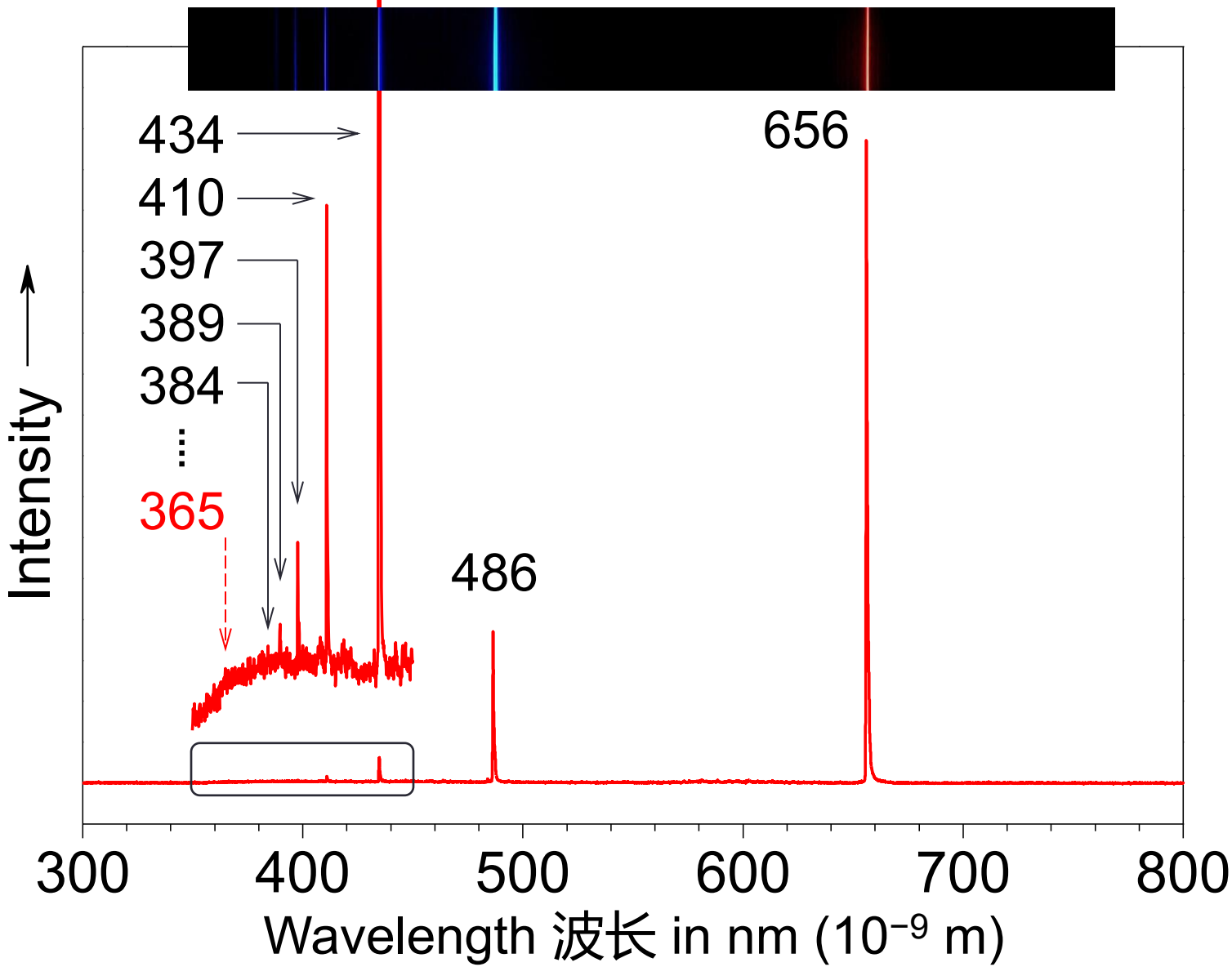


Grating
光栅

Demo: Gas Discharge Spectra



Experimental H Spectrum (1860s)



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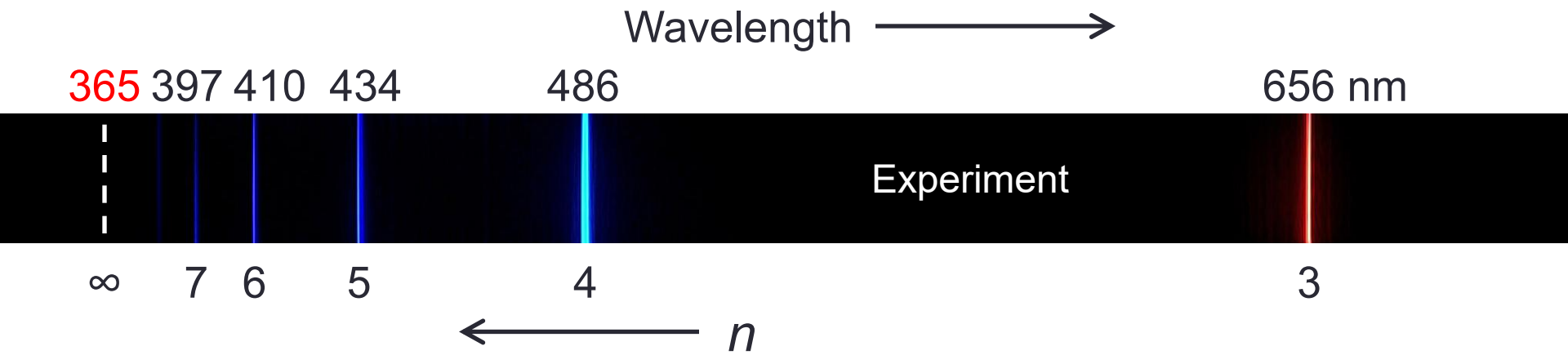
The Balmer Formula (1885)



$$\frac{656}{365} \approx 1.80 = \frac{9}{5} = \frac{9}{9-4}$$

$$\frac{486}{365} \approx 1.33 = \frac{4}{3} = \frac{16}{16-4}$$

$$\Longrightarrow \lambda_n = B\frac{n^2}{n^2-4}, \quad B = 365 \text{ nm}, \quad n = 3,4,5\ldots$$

The Balmer Formula (1885)

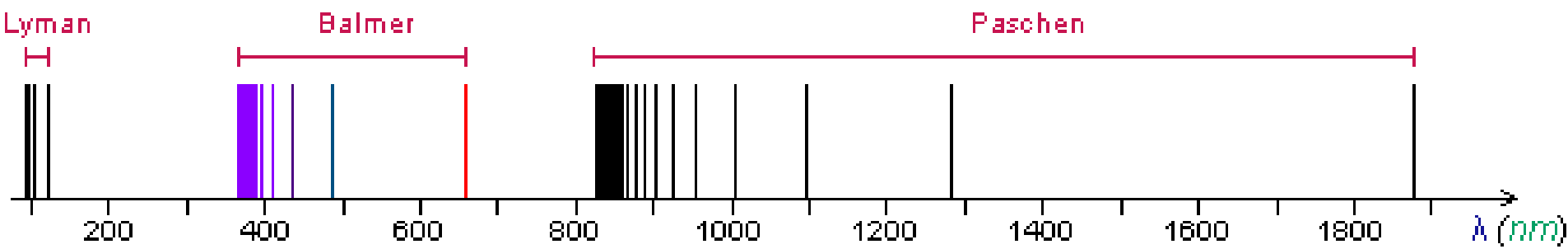
TABLE II.—BALMER'S SERIES OF LINES IN THE ARC SPECTRUM OF HYDROGEN.

H line.	n	Wave-length, λ. in Å (10^{-10} m) Calculated.	Observed.	Difference.
H_α or C	3	6564·96	6564·97	+0·01
H_β or F	4	4862·93	4862·93	—
H_γ or G	5	4341·90	4342·00	−0 1
H_δ or h	6	4103·10	4103·11	+0·01
H_ϵ or H	7	3971·4	3971·4	—
H_ζ or α	8	3890·3	3890·3	—
H_η or β	9	3836·7	3836·8	+0·1
H_θ or γ	10	3899·2	3799·2	—
H_ι or δ	11	3771·9	3771·9	—
H_κ or ϵ	12	3751·4	3751·3	−0·1
H_λ or ζ	13	3735·6	3735·3	−0·3
H_μ or η	14	3723·2	3722·8	−0·4
H_ν or θ	15	3713·2	3712·9	−0·3

1. J. S. Ames, *Philosophical Magazine*, **1890**, *30*, 33.

The Rydberg Formula (1888)

$$\lambda_n \propto \frac{n^2}{n^2 - 1} \qquad \lambda_n \propto \frac{n^2}{n^2 - 4} \qquad \lambda_n \propto \frac{n^2}{n^2 - 9}$$



$$\lambda_n = B\frac{n^2}{n^2-4}$$

$$\Longrightarrow \quad \frac{1}{\lambda_n} = \frac{1}{B}\left(1 - \frac{4}{n^2}\right)$$

$$\Longrightarrow \quad \frac{1}{\lambda_n} = \frac{4}{B}\left(\frac{1}{2^2} - \frac{1}{n^2}\right)$$

$$\frac{1}{\lambda} = R_H\left(\frac{1}{m^2} - \frac{1}{n^2}\right),\ m < n.$$

R_H: Rydberg's constant

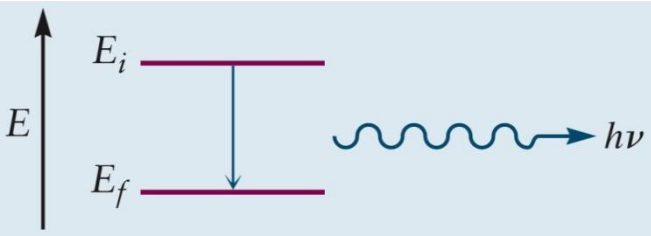
Bohr's Interpretation (1913)

$$\frac{1}{\lambda} = R_H\left(\frac{1}{m^2} - \frac{1}{n^2}\right),\ m < n.$$

$$h\nu = \Delta E = E_{final} - E_{initial}$$

Niels Bohr (Copenhagen, Cambridge, 1885–1962)

- In an atom, there are various states 能态 of fixed, discrete energy.
- Emission or absorption of light results from a transition 跃迁 between two of these states.
- The energy quantum $h\nu$ of the absorbed/emitted light equals the transition energy.



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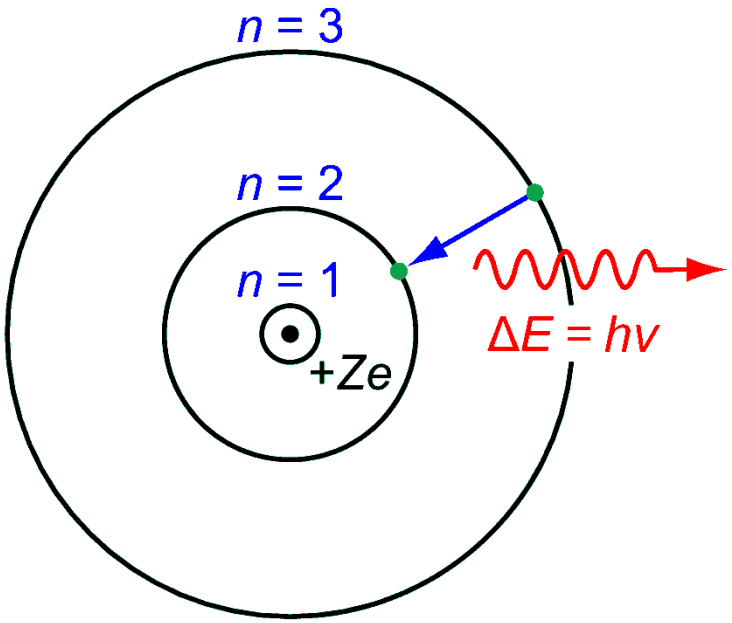
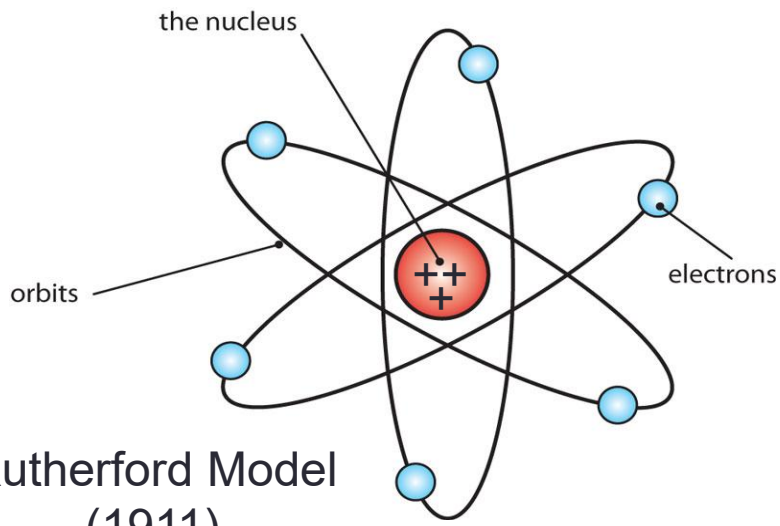
The Bohr Model for H



- The electron orbits the nucleus.
- Each orbit is a circle specified by an angular momentum $L = nh/2\pi$ ($n \in \mathbb{N}$).
- The electron is stable in these orbits, but gains or loses energy when jumping between the orbits.



Rutherford Model (1911)

Shanghai Synchrotron (2009)

The Virial Theorem for Circular Orbits

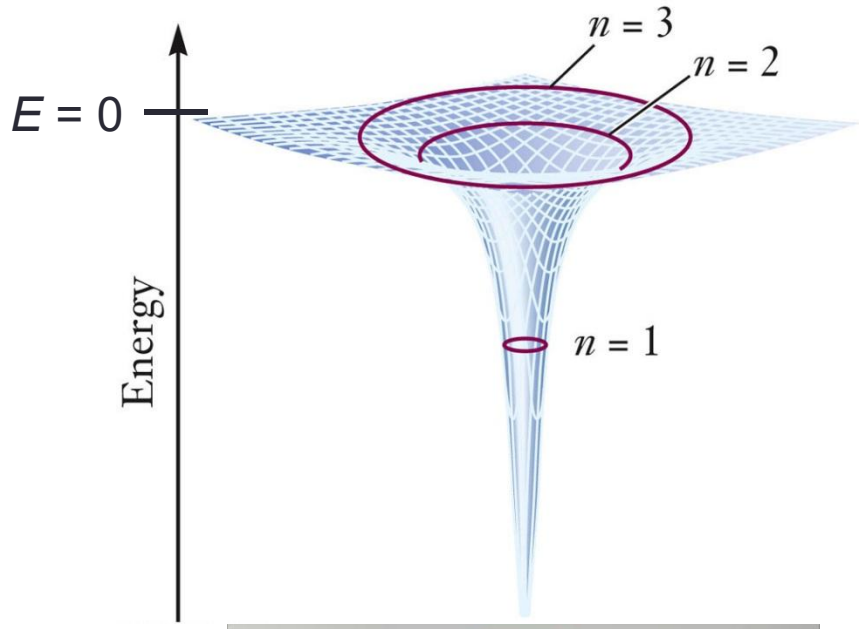
- Coulomb attraction = centripetal force 向心力

$$F = \frac{1}{4\pi\varepsilon_0}\frac{Ze^2}{r^2} = m_e\frac{v^2}{r}$$

$$\Longrightarrow \quad \frac{1}{4\pi\varepsilon_0}\frac{Ze^2}{r} = m_e v^2$$

$$\Longrightarrow \quad \frac{1}{8\pi\varepsilon_0}\frac{Ze^2}{r} = \frac{1}{2}m_e v^2 = E_k$$

Because $V = -\frac{1}{4\pi\varepsilon_0}\frac{Ze^2}{r}$,



$$E_k = -\frac{V}{2}, \quad E = E_k + V = \frac{V}{2}.$$

The Quantization Condition

- Coulomb attraction = centripetal force 向心力

(1) $$F = \frac{1}{4\pi\varepsilon_0}\frac{Ze^2}{r^2} = m_e\frac{v^2}{r}$$

- Discrete orbits

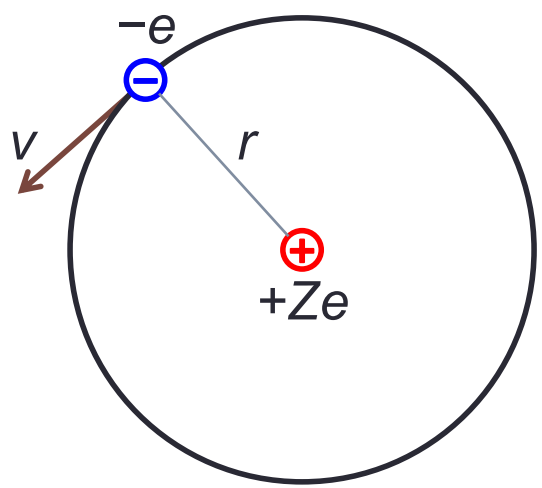
(2) $$E = \frac{V}{2} = -\frac{1}{8\pi\varepsilon_0}\frac{Ze^2}{r}$$

(3) $$E \propto \frac{1}{n^2},\ n \in \mathbb{N}$$

(2)+(3) $\Longrightarrow r \propto n^2$

(1) becomes $F \propto \frac{1}{n^4} \propto \frac{v^2}{n^2} \Longrightarrow v \propto \frac{1}{n}$

Therefore $L = m_e v \cdot r \propto n$



$$L = m_e v r = n\frac{h}{2\pi} = n\hbar,\ n \in \mathbb{N}$$

Solving the Unknowns: *r* and *v*

- Coulomb attraction = centripetal force

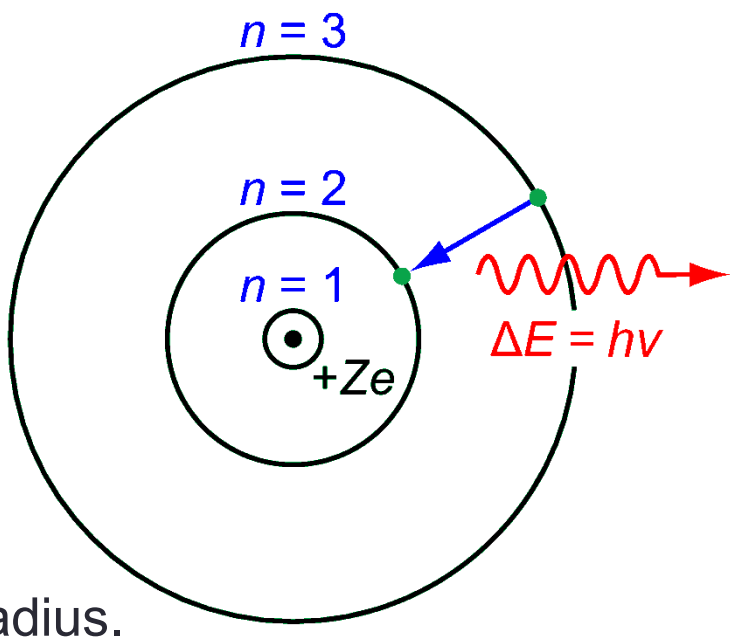
(1) $$F = \frac{1}{4\pi\varepsilon_0}\frac{Ze^2}{r^2} = m_e\frac{v^2}{r}$$

- Quantized angular momentum L

(2) $$L = m_e v \cdot r = n\frac{h}{2\pi},\ n \in \mathbb{N}$$

(1)+(2)

$$r = \frac{n^2}{Z}\frac{\varepsilon_0 h^2}{\pi e^2 m_e} = \frac{n^2}{Z}a_0,$$ a_0 = 52.9 pm is Bohr radius.



For hydrogen atom (Z = 1),

n = 1, $r = a_0$ = 52.9 pm;

n = 2, $r = 4a_0$ = 212 pm;

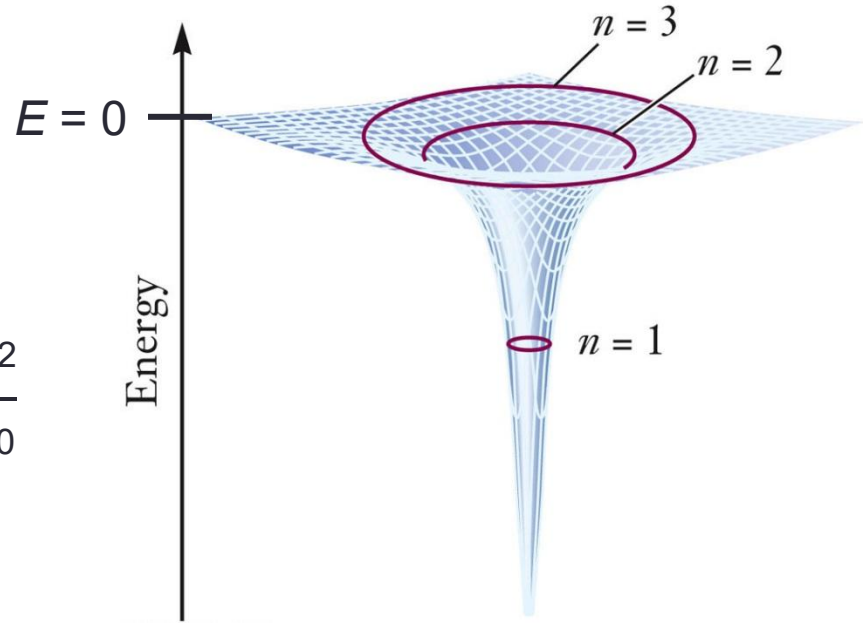
n = 3, $r = 9a_0$ = 476 pm.

$$r = \frac{n^2}{Z}a_0$$

Orbital Energies

From previous pages:

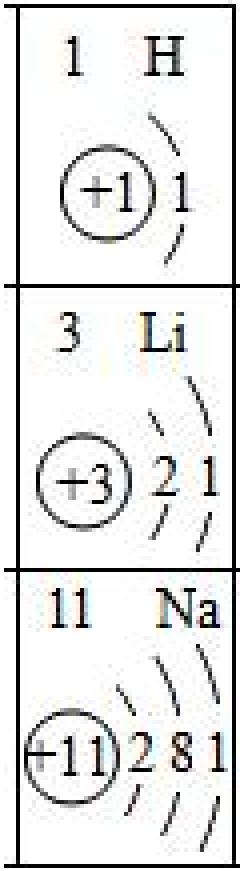
$$\left.\begin{aligned} E &= -\frac{1}{8\pi\varepsilon_0}\frac{Ze^2}{r} \\ r &= \frac{n^2}{Z}a_0 \end{aligned}\right\} \Rightarrow E = -\frac{1}{8\pi\varepsilon_0}\frac{Z^2e^2}{n^2a_0}$$



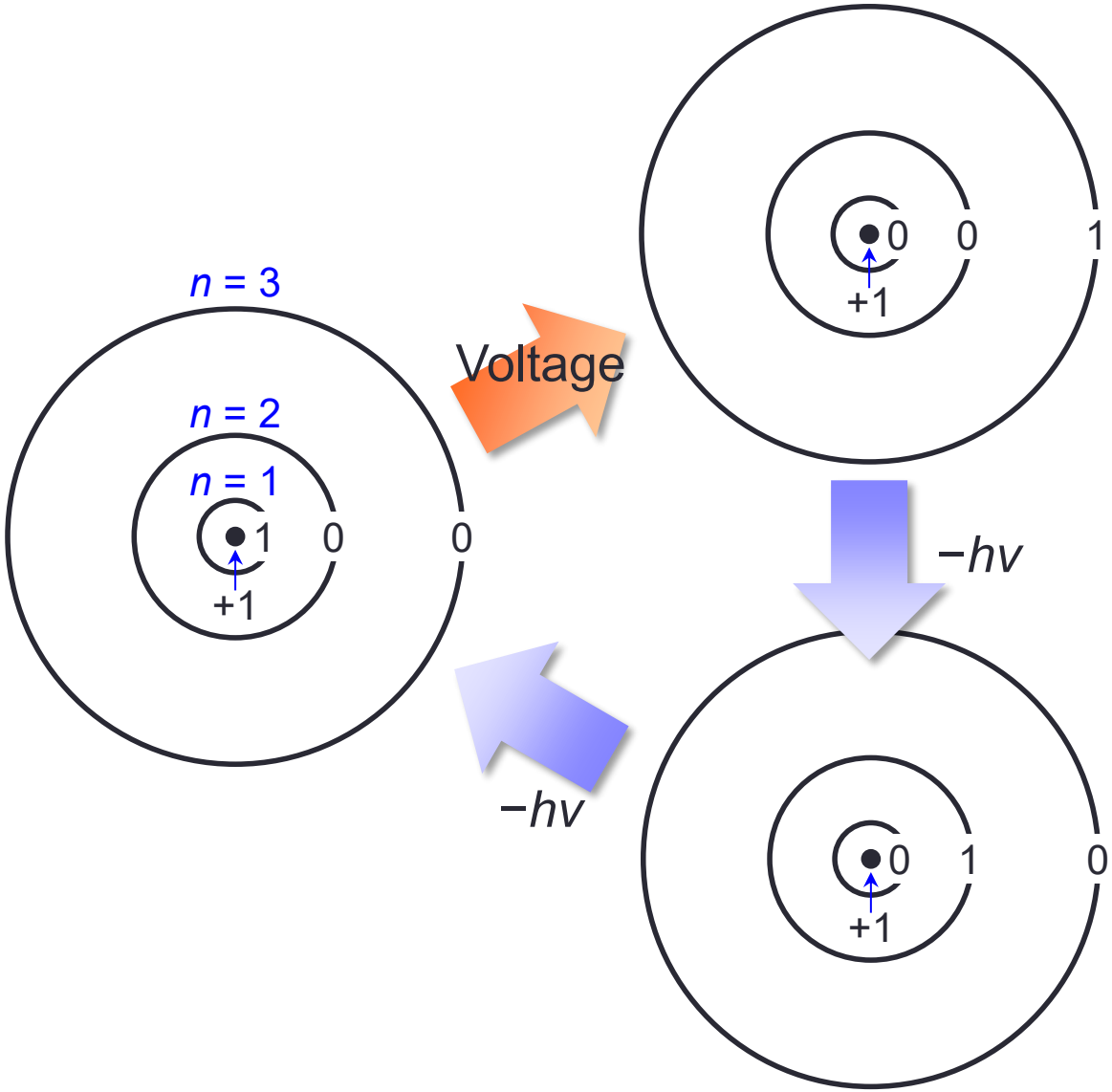
$$E = -\mathrm{Ry}\frac{Z^2}{n^2},\ \mathrm{Ry} = 2.18\times10^{-18}\ \mathrm{J} = 13.6\ \mathrm{eV}$$

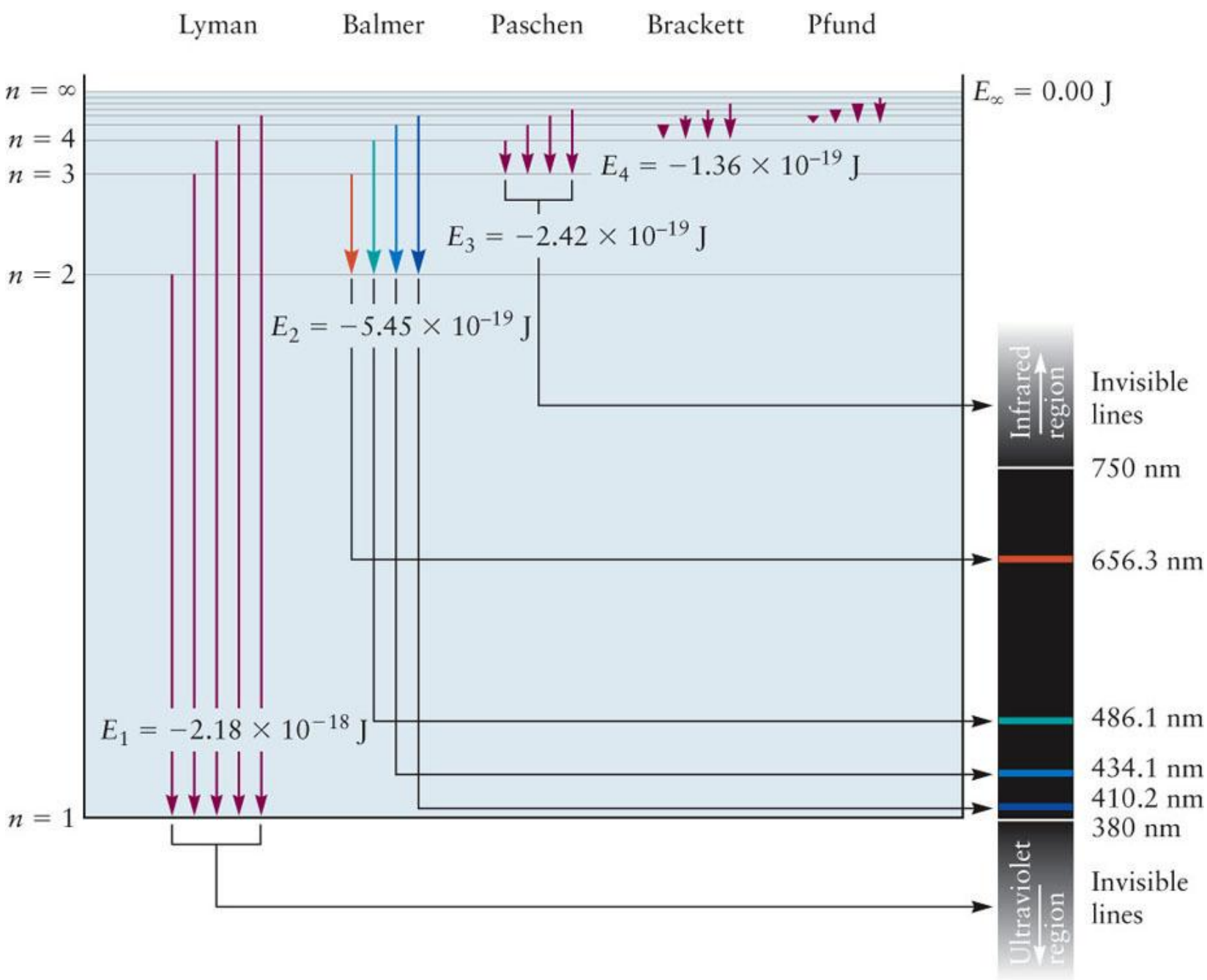
Ry: Rydberg energy

Bohr Model and Electron Orbits



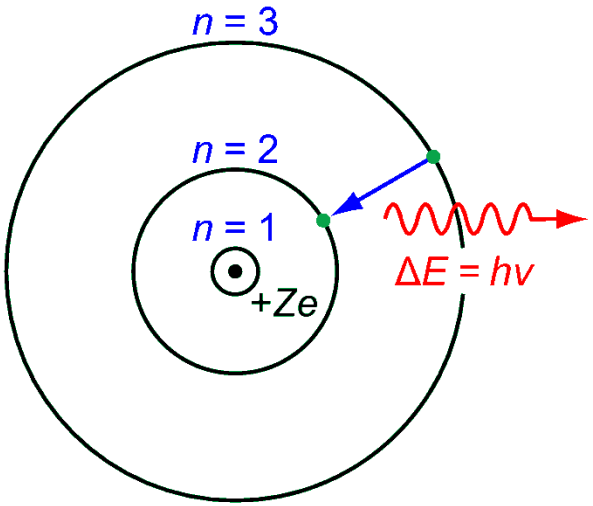
VS




Lyman
Balmer
Paschen
Brackett
Pfund
n = ∞
n = 4
n = 3
n = 2
n = 1
E∞ = 0.00 J
E4 = −1.36 × 10−19 J
E3 = −2.42 × 10−19 J
E2 = −5.45 × 10−19 J
E1 = −2.18 × 10−18 J
Infrared region
Invisible lines
750 nm
656.3 nm
486.1 nm
434.1 nm
410.2 nm
380 nm
Ultraviolet region
Invisible lines

Summary for the Bohr Model

- Stable orbits / states of H atom
- Transitions between the orbits
- Quantized orbital angular momentum
- “Old quantum theory”



But the Bohr model doesn’t work for the He atom at all!

Next lecture series: Quantum Mechanics

Reading: OGB8 §4